Evaluating and Comparing Confirmatory Factor Models

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Chapter box

Five themes.
Global fit is a profile, not a verdict. The exact-fit test, RMSEA, CFI, TLI and SRMR answer different questions with different references, and a favorable profile is compatible with a model that is wrong in a specific, locatable way.
Discrepancy has a location. Residual matrices and one-parameter diagnostics say where a model strains; a single summary cannot.
Comparison requires eligibility. Nested tests and information criteria apply only when their assumptions and likelihood bases make the comparison meaningful; printing two rows in a table is not a test.
Equivalent models fit identically. With three first-order factors, the correlated and higher-order representations are reparameterizations of one another, and no amount of data can choose between them.
A revision needs evidence it did not create. A change suggested by a sample is a hypothesis about the population, and only records not used to suggest it can support it.
Prerequisites. The covariance equation, restrictions, identification, ML estimation, parameter metrics and uncertainty developed earlier, and the checked fitted object they produced.
Records. The twelve-indicator fit on the confirmation sample, a fixed validation sample used once, and a separate six-indicator example for compact comparisons.
Scope. We evaluate, compare, extend, revise and check against the known generating population. Fit does not establish construct reality, measurement invariance, or causal direction.

1 From an estimated model to an evaluation record

1.1 Begin with what is already fixed

Chapter 3 supplied a converged, admissible estimate of M1-cross on Sample B. The candidate was fixed using Sample A: twelve continuous indicators, three correlated factors, P4 additionally loading on Strategy, and S4 additionally loading on Belonging. All other cross-loadings and all residual covariances are zero. Factor variances equal one, and the primary model is covariance-only. It has 29 free parameters and 49 df.

This chapter refits that exact specification to Sample B, taken from the released file. It does not reconstruct the baseline from rounded estimates, change the item order, or substitute a simple-structure model, so the fitted object below carries the same parameter table, software settings, warnings and sample covariance under the normal-likelihood convention.

```r
sle <- read.csv("sle_continuous.csv")

items <- c(paste0("P", 1:4),
           paste0("S", 1:4),
           paste0("B", 1:4))

dat <- sle[sle$sample == "B", items]

model <- '
  Planning  =~ p1*P1 + p2*P2 + P3 + P4
  Strategy  =~ S1 + S2 + S3 + S4 + P4
  Belonging =~ B1 + B2 + B3 + B4 + S4
'
fit <- lavaan::cfa(model, data = dat, estimator = "ML",
                   likelihood = "normal", std.lv = TRUE,
                   meanstructure = FALSE)
lavaan::fitMeasures(fit, c("chisq", "df", "pvalue", "rmsea",
                           "cfi", "tli", "srmr"))
```

This is the same specification the previous chapter estimated, refitted from the released file so that the block runs as printed. It is not a license to skip checking the result. Before any fit statistic below is used, the render confirms that the released records match the declared population, that the item order and free-parameter count are as specified, and that the solution converged and is admissible. The R and package versions are recorded alongside those checks so that a later reader can tell what produced the numbers; they are not themselves verified against a required version.

1.2 Separate five questions

Question	Evidence	What a favorable answer does not establish
Did the optimizer finish?	Convergence, gradients, warnings	Admissibility or adequacy
Is the solution admissible?	Variances, covariance matrices, local rank	Small model discrepancy
How large is global discrepancy?	Model test and fit profile	Absence of important local problems
Where are discrepancies located?	Residual patterns and parameter diagnostics	A unique explanation

Question	Evidence	What a favorable answer does not establish
Is interpretation defensible?	Content, estimates, uncertainty, analytic history	Validity for every score use

These questions are related but not interchangeable. An optimizer can converge to a badly fitting model. A globally favorable fit profile can coexist with a questionable item or weakly identified parameter. A well-measured factor can still be used in an inappropriate substantive analysis.

The conventional output order often begins with the test statistic, but our reading order starts with the model and estimation contract. Fit measures belong to a specified analysis, not to the data set in isolation.

1.3 Keep the evidence roles visible

Sample A developed the loading candidate. Sample B estimates and evaluates that fixed candidate and its planned rivals. A new relation suggested by B is exploratory in B, even if only one path is changed. Sample C can assess that relation independently only after the changed specification is fixed. All three samples have 360 cases from the same documented synthetic continuous population; no new seed is selected to obtain a preferred teaching outcome.

The six-indicator simulated example is separate. Its 500 newly generated continuous records illustrate four model specifications and transparent likelihood comparisons. They are drawn from a different population than the SLE samples and cannot borrow those samples' development or confirmation status.

2 What estimation is fitting: one continuous ML family

2.1 Reconstruct the discrepancy before interpreting it

Write the implied covariance as

$$\Sigma(\vartheta) = \Lambda\Phi\Lambda^{\mathsf{T}} + \Theta.$$

For p continuous outcomes and the declared sample covariance,

$$F_{\mathrm{ML}} = \log|\Sigma(\vartheta)| + \mathrm{tr}\{\mathbf{S}\Sigma(\vartheta)^{-1}\} - \log|\mathbf{S}| - p.$$

The first two terms depend on the model. The remaining terms make the discrepancy zero when the implied covariance equals the observed one. The fitted value is the minimum over the specified parameter space, not necessarily a small number by any substantive standard.

In our normal-likelihood analysis, $\mathbf{S} = \mathbf{S}_N$ uses divisor N, and the model statistic is

$$T_M = N\widehat{F}_{\mathrm{ML}}.$$

For M1-cross, the reconstructed discrepancy is 0.134653; multiplying by 360 gives 48.475. The calculation agrees with the stored software statistic. This check catches mismatched divisors, a wrongly ordered matrix, or an incorrect dimension term before those mistakes propagate into fit-index calculations.

```
S <- lavaan::lavInspect(fit, "sampstat")$cov
Sigma <- lavaan::fitted(fit)$cov
p <- ncol(S)
Fml <- determinant(Sigma, logarithm = TRUE)$modulus +
       sum(diag(solve(Sigma, S))) -
       determinant(S, logarithm = TRUE)$modulus - p
as.numeric(nrow(dat) * Fml)
```

2.2 Do not mix normal, Wishart, and robust conventions

R's ordinary sample covariance uses divisor $N-1$. With raw-data normal ML, `lavaan` handles conversion to divisor N internally. Wishart ML uses an $N-1$ covariance and test multiplier. The official estimator documentation explains this distinction. A small difference between test statistics can be a convention difference rather than a scientific disagreement.

Changing an observed unit can leave ML discrepancy unchanged when the entire model and all restrictions are transformed consistently. Leaving a raw-loading equality unchanged after changing an item's unit is not such a transformation. Scale invariance is conditional on comparing equivalent model specifications.

The present calculations use conventional ML throughout. Later robust ML procedures adjust standard errors and tests for particular forms of nonnormality; one cannot subtract two displayed scaled statistics and assume an ordinary difference test. FIML later handles missing continuous observations under stated missingness and model assumptions. Neither topic supplies a categorical-indicator measurement model, which is outside this series' present scope.

2.3 Large-sample properties are conditional

Consistency concerns increasing sample size under identification and other appropriate conditions, not merely increasing the number of indicators. Asymptotic normality of estimates and chi-square references are approximations, not guarantees for every finite sample or boundary configuration. More indicators can add information, but also add parameters, content heterogeneity, and possible misspecification.

Different starts need not always converge to the same optimum. Negative residual variances, singular factor covariance matrices, or very large standard errors call for investigation. Here all planned SLE and example-model fits pass the recorded convergence and admissibility gates. That makes ordinary evaluation interpretable; it does not predetermine the evaluation's conclusion.

3 Read global fit as a profile

3.1 The exact-fit null

The covariance-only null is that there exists an admissible parameter vector such that

$$H_0 : \Sigma_{\text{population}} = \Sigma(\vartheta).$$

Under regularity and the conventional ML assumptions, $T_M \mathrel{\dot\sim} \chi^2_{df_M}$ when that null holds. The dot marks an asymptotic approximation. The df refer to distinct covariance restrictions after identification and parameter accounting, not N minus the number of regression coefficients.

For M1-cross, $T_M = 48.475$ on 49 df, with $p = 0.494$. We fail to reject exact fit under this test. We do not "accept the true model." Sampling variability, power, alternative covariance-equivalent models, and departures not well detected by the test remain relevant.

Large samples can detect small fixed discrepancies. That is not a reason to declare the test untrustworthy solely because N is large; it is a reason to separate evidence against *exact* fit from the magnitude and consequences of the discrepancy. Conversely, a small sample can fail to reject a consequential departure.

3.2 RMSEA: discrepancy per degree of freedom

The root mean square error of approximation (Steiger & Lind, 1980) measures discrepancy per degree of freedom rather than total discrepancy, which is what makes it comparable across models of different size. For the present N convention,

$$\widehat{\mathrm{RMSEA}} = \sqrt{\max\left\{\frac{T_M - df_M}{N\,df_M}, 0\right\}}.$$

Under the commonly taught $N-1$ convention, replace N by $N-1$. State which calculation is being used. The numerator estimates a noncentrality contribution; it is truncated at zero when $T_M \le df_M$. Thus RMSEA equal to zero does not require zero residuals or $T_M = 0$.

The interval is obtained by inverting a noncentral chi-square reference and transforming its noncentrality limits onto the RMSEA scale. A 90% interval is conventional in this literature, whereas the parameter intervals in Chapter 3 were 95%. These different levels should be labeled, not silently mixed.

For M1-cross, RMSEA is 0.000, with 90% interval [0.000, 0.034]. The zero estimate arises because the test statistic is slightly below its df. The nonzero upper limit expresses uncertainty about population approximation error.

Low-df models can have unstable RMSEA estimates and broad intervals. A point estimate without its interval can conceal that uncertainty. Nor does an interval imply equal plausibility of every value inside it. Explicit close-fit or not-close-fit hypotheses require stated population thresholds and their own tests; a threshold should not be inferred retroactively from whichever endpoint looks favorable. The interval and power perspective is developed by MacCallum et al. (1996).

3.3 CFI and TLI need a baseline

Incremental indices compare a model against a reference rather than against exact fit, so the reference has to be stated before the number means anything (Hu & Bentler, 1999). For continuous covariance analysis, the usual independence baseline frees the observed variances and fixes covariances to zero. With twelve indicators, it has 66 covariance restrictions. It is not a competing substantive theory of the three factors; it supplies a reference for incremental improvement.

One conventional CFI expression is

$$\widehat{\mathrm{CFI}} = 1 - \frac{\max(T_M - df_M, 0)}{\max(T_M - df_M, T_B - df_B, 0)}.$$

If the denominator is zero, this expression requires special handling; it is not an ordinary ratio with a meaningful zero denominator. In our record the baseline statistic is 1594.445 on 66 df, so that issue does not arise. A high CFI means the model improves substantially relative to this baseline under the stated correction, not that the construct interpretation has been established.

For positive baseline discrepancy, TLI can be written

$$\widehat{\mathrm{TLI}} = \frac{T_B/df_B - T_M/df_M}{T_B/df_B - 1}.$$

Unlike a correlation coefficient, this formula need not remain between zero and one. A value slightly above one is possible when $T_M < df_M$; it is not "better than perfect measurement." We retain the software's uncapped conventional value rather than altering it to look more familiar.

Baseline dependence explains why a model with substantial local discrepancy can still have a high incremental index. The baseline may be much worse. An index's favorable appearance must be interpreted relative to the question it answers.

3.4 Compare the planned profiles

Table 1

Sample B planned models, all using identical observations, indicators, and conventional normal ML. M1-cross is the inherited target.

Model	T	df	p	RM-SEA	CI90	CFI	TLI	SRMR
M0_simple	135.142	51	<.001	0.068	[0.054, 0.082]	0.945	0.929	0.063
M1_cross	48.475	49	0.494	0.000	[0.000, 0.034]	1.000	1.000	0.032
M2_orthogo-nal	121.189	52	<.001	0.061	[0.047, 0.075]	0.955	0.943	0.139
M3_equal	54.948	50	0.293	0.017	[0.000, 0.039]	0.997	0.996	0.039
M4_one	695.266	54	<.001	0.182	[0.170, 0.194]	0.580	0.487	0.134

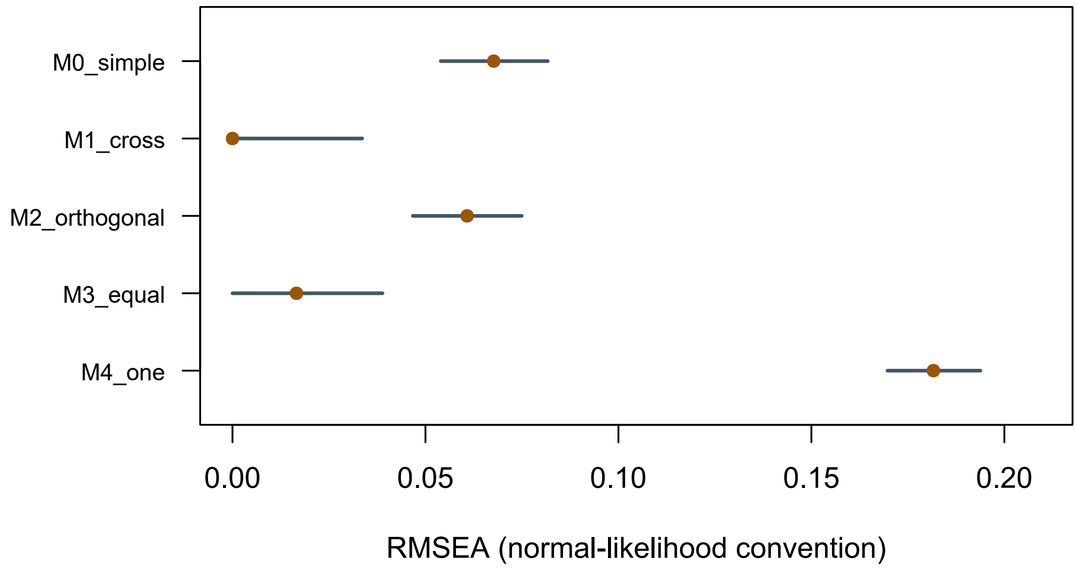

Figure 1: RMSEA point estimates and 90 percent intervals for the planned models. No traffic-light cutoffs are imposed.

The profiles do not reduce to unanimous votes. M2-orthogonal has CFI about .955 but SRMR about .139: appreciable standardized residual discrepancy remains despite a sizable improvement over independence. M3-equal has a favorable global profile even though the targeted loading-equality test is opposed. Global approximation and a focused parameter restriction are different questions.

M1-cross has the most favorable overall balance among these prespecified alternatives. M0-simple leaves more discrepancy after forcing the two secondary loadings to zero. M4-one does not provide a satisfactory broad covariance representation relative to these multi-factor alternatives. These are judgments about this synthetic sample and candidate set, not universal thresholds for instrument validation.

4 Locate local discrepancy

4.1 Three residual scales

The raw covariance residual is

$$e_{jk} = s_{jk} - \hat{\sigma}_{jk}.$$

Its unit is the product of the two indicators' units. A residual of 10 has a different meaning for indicators with SDs 2 and 3 than for indicators with SDs 10 and 12. Two useful rescalings are

$$e_{jk}^{(s)} = \frac{s_{jk} - \widehat{\sigma}_{jk}}{\sqrt{s_{jj}s_{kk}}}, \qquad e_{jk}^{(r)} = r_{jk} - \widehat{r}_{jk}.$$

The first divides both observed and fitted covariance by *observed* SDs. The second separately converts each covariance matrix to a correlation matrix. They coincide off the diagonal when the observed and fitted variances match; otherwise they can differ. Neither is a residual divided by its sampling standard error. A standardized residual in the variance-scaling sense is not automatically a z test.

For this covariance-only ML model with free residual variances, fitted variances closely reproduce observed variances, so the two off-diagonal rescalings are numerically very close. The script nevertheless calculates them separately and checks the actual relationship rather than assuming it in every future model.

4.2 SRMR is a root mean square, not a mean absolute residual

For the covariance-based convention verified here,

$$\mathrm{SRMR} = \sqrt{\frac{2}{p(p+1)} \sum_{j=1}^{p} \sum_{k=1}^{j} \left(e_{jk}^{(s)}\right)^2}.$$

The average includes the diagonal and one triangle. Squaring gives large discrepancies extra influence, and the square root returns the scale to standardized covariance units. This differs from the mean absolute residual and from Chapter 2's off-diagonal-only RMS.

The direct calculation gives 0.032, matching the package SRMR. A small summary does not imply every cell is small: many near-zero cells can dilute a few larger ones. Mean structures, multiple groups, and alternative estimators require their own checked conventions.

4.3 Read patterns in the twelve-item system

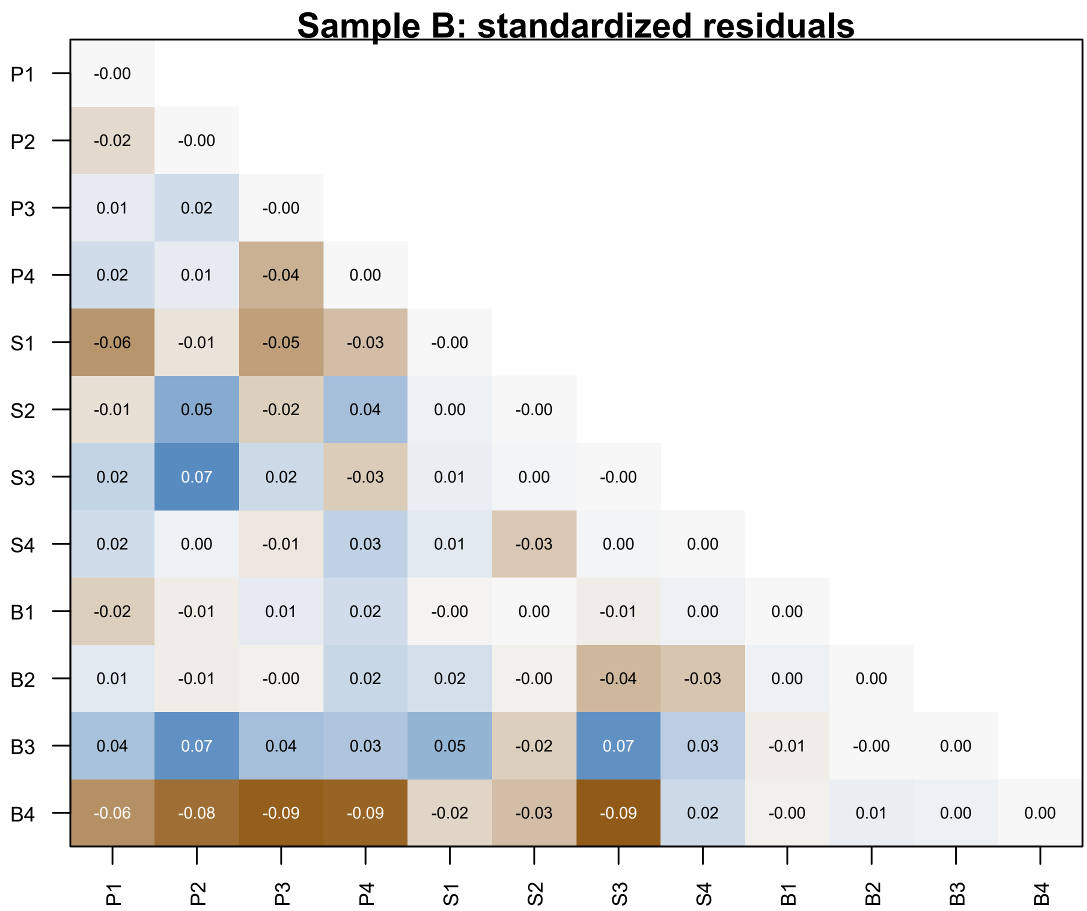

Figure 2: M1-cross standardized covariance residuals in Sample B, lower triangle including the diagonal. Numeric cells retain the sign of observed minus implied covariance.

Table 2
Eight largest absolute standardized covariance residuals. Raw and correlation residuals are shown separately.

Item1	Item2	Observed	Implied	Raw	Std	Correlation
B4	S3	-0.371	10.660	-11.031	-0.093	-0.093
B4	P3	-0.270	11.407	-11.677	-0.092	-0.092
B4	P4	2.598	10.883	-8.285	-0.088	-0.088
B4	P2	1.789	10.597	-8.809	-0.082	-0.082
S3	P2	33.518	24.903	8.615	0.072	0.072
B3	P2	16.920	11.118	5.802	0.070	0.070
B3	S3	17.477	11.183	6.293	0.069	0.069
B4	P1	3.409	8.554	-5.145	-0.061	-0.061

For B4 and S3, the observed covariance is -0.371 and the implied covariance is 10.660. Their raw residual is -11.031; scaled by the observed SDs it is -0.093. The model overpredicts this pair's covariance. Several other negative cross-domain residuals also involve B4, so it is reasonable to inspect that item rather than treat the largest cell as an isolated software signal.

B4 concerns connection to the institution as a whole, whereas B1–B3 refer more directly to peers, participation, and classroom voice. That distinction suggests questions about content breadth and local heterogeneity. It does not uniquely imply that B4 should load negatively on Planning. A cross-loading, an omitted narrower factor, a response process difference, and sampling fluctuation can leave overlapping residual patterns.

Before modifying anything, check coding, distributions, data dependence, and whether the proposed relation changes the intended measurement meaning. A globally favorable fit does not prohibit local investigation, but it also does not require a modification merely to make every residual smaller.

5 Compare planned rivals and test constraints

5.1 Establish nesting by exact restrictions

For compatible conventional ML fits, if restricted model R is regularly nested in unrestricted model U,

$$\Delta T = T_R - T_U, \qquad \Delta df = df_R - df_U, \qquad \Delta T \mathrel{\dot\sim} \chi^2_{\Delta df}$$

under the null restrictions and regularity conditions. The null is a population parameter restriction, not that the two observed fit statistics are literally equal. The larger model cannot have a worse optimized likelihood if both optimizations have found the corresponding optima. That monotonicity alone does not establish the chi-square reference.

For M0 versus M1, set the P4-on-Strategy and S4-on-Belonging loadings to zero. For M2 versus M1, set all three factor correlations to zero. For M3 versus M1, impose the one raw-loading equality. These are different hypotheses with different df changes.

```
lavaan::lavTestLRT(fit_M1, fit_M0, method = "standard")
```

The compatible fits must use identical indicators, cases, data scale, mean structure, likelihood convention, and sampling assumptions. A smaller free-parameter count is not proof of nesting. The one-factor model's relation to a three-factor model passes through degenerate latent covariance configurations, so we do not assign an ordinary chi-square difference test to M4 merely by subtracting df.

Table 3

Eligible planned restrictions tested against M1 in Sample B. Values reproduce lavTestLRT under conventional ML.

Comparison	Delta T	Delta df	p
M0_vs_M1	86.667	2	<.001
M2_vs_M1	72.714	3	<.001
M3_vs_M1	6.473	1	0.011

The two zero cross-loading restrictions cost 86.667 chi-square units on two df. The orthogonality restrictions cost 72.714 on three df. Under this family, both are strongly opposed. The raw-loading equality costs 6.473 on one df, with $p = 0.011$. Its relatively favorable global fit profile did not make that particular restriction innocuous.

Failure to reject a restriction would not establish equivalence. It would leave the data insufficient to reject it at the chosen level under that test. If practical equivalence is the question, specify a meaningful discrepancy region and an appropriate procedure rather than translate nonsignificance into equality.

5.2 Four planned models on the six-indicator example

The separate example specifies the following models before the six-indicator example is generated. All use fixed-factor scaling, free residual variances, diagonal residual covariance, and conventional normal ML.

Model	F1 indicators	F2 indicators	Factor relation	Free / df
D1	Y1, Y2, Y3	Y4, Y5, Y6	Correlated	13 / 8
D2	Y1, Y2	Y3, Y4, Y5, Y6	Correlated	13 / 8
D3	Y1, Y2, Y3	Y3, Y4, Y5, Y6	Correlated	14 / 7
D4	Y1, Y2, Y3	Y3, Y4, Y5, Y6	Orthogonal	13 / 8

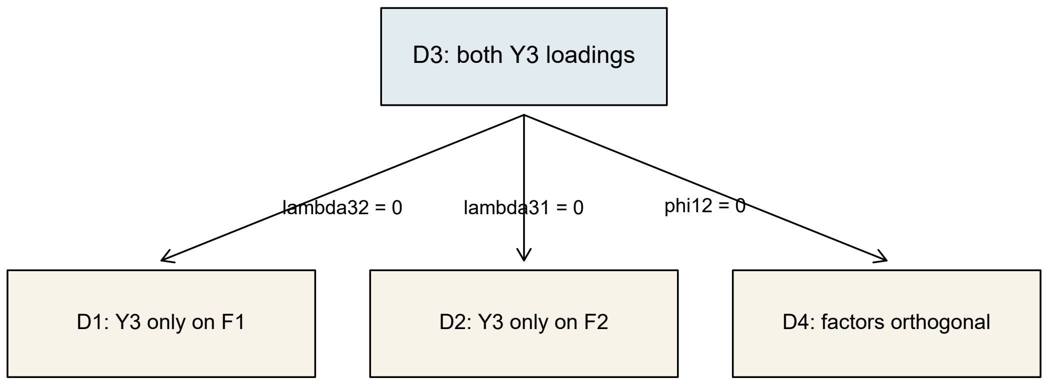

Figure 3: Three different one-parameter restrictions of D3. D1 and D2 are not nested in one another merely because both have eight df.

Table 4

Recomputed fit profiles for the separate simulated six-indicator example, N = 500.

Model	T	df	p	RMSEA	CI90	CFI	TLI	SRMR
D1	42.463	8	<.001	0.093	[0.067, 0.121]	0.974	0.952	0.053
D2	334.998	8	<.001	0.286	[0.260, 0.312]	0.756	0.542	0.173
D3	8.469	7	0.293	0.020	[0.000, 0.061]	0.999	0.998	0.016
D4	17.980	8	0.021	0.050	[0.018, 0.081]	0.993	0.986	0.068

D1 has CFI about .974 but RMSEA about .093, with a 90% upper limit above .12. This is another reason not to label models by one index. D3's additional loading changes the covariance representation and substantially reduces discrepancy. The D1–D3 difference is 33.994 on one df. Its result is computed for this new six-indicator example; it is not inherited from an older example with different data.

The two-indicator F1 block in D2 is not isolated: it is linked through a freely estimated nonzero factor covariance to a factor with several indicators. That distinguishes it from Chapter 3's orthogonal isolated- block counterexample. The actual numerical fits pass the local and admissibility checks. A generic "two indicators always fail" rule would miss the role of the rest of the model.

5.3 Score, Wald, and LR: three views of one restriction

In D2, $\lambda_{31} = 0$: Y3 has no F1 loading. D3 frees it. The score test uses the restricted fit and asks whether the likelihood has a slope suggesting that release. A modification index is the corresponding local one-parameter score statistic. The Wald test uses the unrestricted estimate and its uncertainty. The LR test fits both and compares their optimized likelihoods.

Table 5

Three tests of the same D2-to-D3 loading restriction. All use one df, but their finite-sample numerical values differ.

Test	Statistic	df
Score (MI in D2)	231.629	1
Wald (D3 loading)	360.919	1
LR (D2 versus D3)	326.529	1

The numbers are not interchangeable. Here the omitted loading is large enough that the local score approximation and unrestricted Wald approximation differ appreciably from the actual LR change. Their asymptotic relationship under regular local conditions does not require exact numerical agreement in every finite example.

For equality constraints, use an appropriate score-test routine such as `lavTestScore()` rather than assuming the ordinary zero-parameter MI table tests every labeled equality. The chapter's analysis does not request MIs for the equality-constrained models, avoiding that misinterpretation.

5.4 AIC and BIC compare eligible likelihoods

For log likelihood $\ell(\hat{\vartheta})$ and the software's effective parameter count q,

$$\text{AIC} = -2\ell + 2q, \qquad \text{BIC} = -2\ell + q \log N.$$

Smaller values indicate a relative preference within the eligible set under each criterion's tradeoff. Neither is a model-test p value, an absolute adequacy index, or a guarantee that the preferred candidate is scientifically meaningful. AIC and BIC can disagree because they penalize complexity differently.

Table 6

Likelihood and information criteria for planned Sample B models. Compare rows within this table, not against another sample or the six-indicator example.

Model	logl	npar	aic	bic
M0_simple	-15249.32	27	30552.64	30657.56
M1_cross	-15205.98	29	30469.97	30582.67
M2_orthogonal	-15242.34	26	30536.68	30637.72
M3_equal	-15209.22	28	30474.44	30583.25
M4_one	-15529.38	24	31106.76	31200.03

For M1, the package count is 29. Substituting its log likelihood and $N = 360$ reproduces both criteria. The same observations and outcomes make these rows comparable; the one-factor model can enter this relative likelihood comparison without being assigned a regular nested LR reference.

D1 and D2 provide a compact nonnested comparison with the same six outcomes and 500 cases. Their AICs are 14135.81 and 14428.35, respectively. D1 has the smaller value, but D3's profile still matters in the larger candidate set. Being preferable to a poor alternative does not establish adequate fit.

Do not compare AIC/BIC across A, B, and C, across item subsets or parcels, or across incompatible likelihood definitions. Do not carry an EFA statistic with the same printed name into this table without checking its exact definition. Eligibility precedes arithmetic.

5.5 Loading equality is not full parallelism

M3 tests a raw-loading equality for P1 and P2. As a separate teaching sensitivity, V1 constrains their *residual variances* equal while leaving their loadings free. V1 is not retroactively listed as an A-fixed rival; it illustrates a different measurement restriction. Its LR difference from M1 is 9.158 on one df, with $p = 0.002$.

These restrictions concern different aspects of the model and are scale-dependent. Equal loadings plus equal residual variances are still not a complete statement about parallel measurements when intercepts and means matter. Equal intercepts and the appropriate common true-score structure must also be considered. Nonsignificant tests would not establish exact parallelism, and a loading equality for just one pair would not establish it for an entire instrument.

6 Extensions of the measurement model

6.1 Ask what accounts for the factor correlations

M1-cross represents Planning, Strategy, and Belonging as three correlated first-order factors. That representation estimates their associations without claiming what produces them. A higher-order model asks a more restrictive question: can one additional latent variable reproduce the covariance among those first-order factors?

Let Λ_l contain the item loadings on the first-order factors. A one-factor higher-order model writes

$$\eta_l = \gamma G + \delta, \qquad \Phi_l = \gamma\gamma^{\mathsf{T}} + \Psi_h, \tag{1}$$

where G has unit variance for convenience and Ψ_h is diagonal in this version. Substitution into the measurement covariance gives

$$\Sigma = \Lambda_l(\gamma\gamma^{\mathsf{T}} + \Psi_h)\Lambda_l^{\mathsf{T}} + \Theta. \tag{2}$$

The components of δ are first-order *disturbances*: the parts of Planning, Strategy, and Belonging not represented by G. They are not indicator residuals. The diagram retains M1-cross's two fixed secondary item loadings; changing those while adding G would confound two model changes.

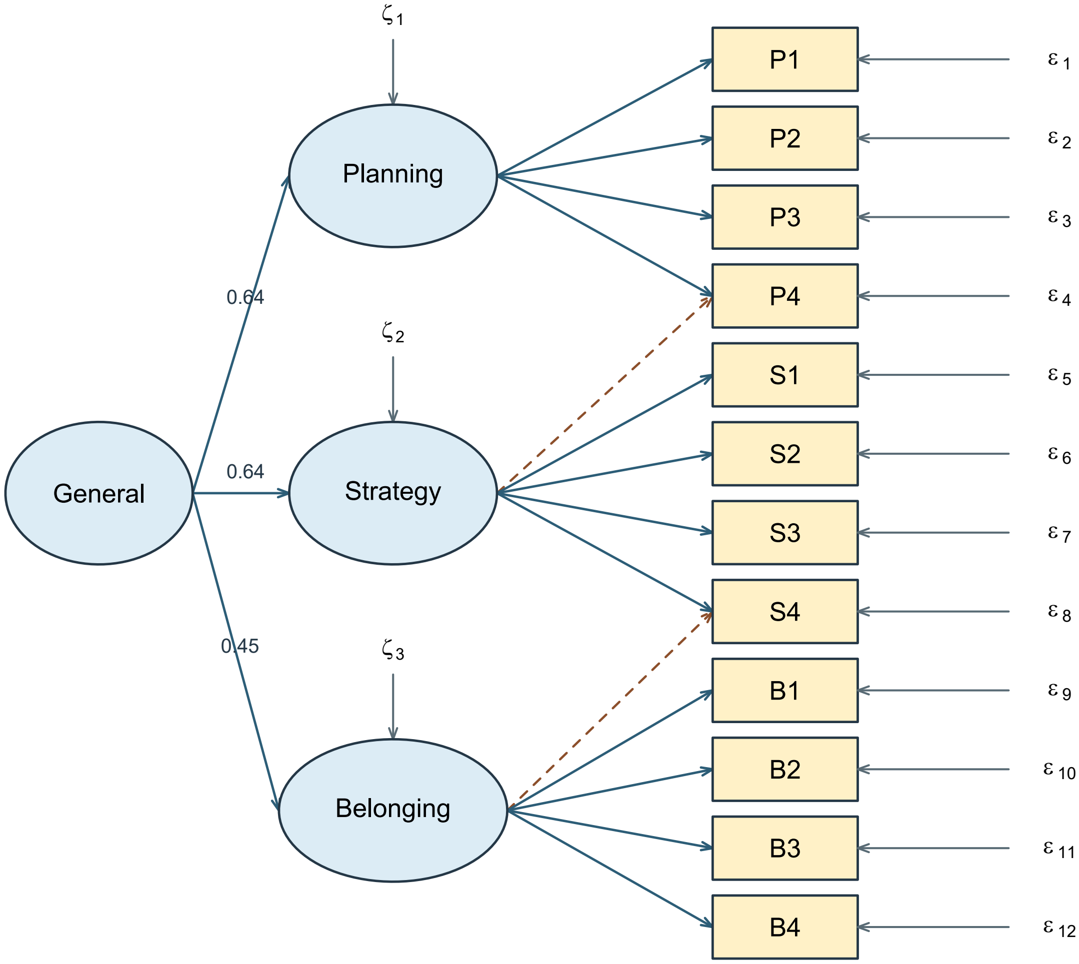

Figure 4: Higher-order representation of the inherited SLE measurement model. General accounts for first-order factor covariances; first-order disturbances remain, and the P4-on-Strategy and S4-on-Belonging loadings are retained.

An arrow from G to a first-order factor represents a hypothesized common source of covariance. Cross-sectional covariance data do not establish that causal direction. Nor does a favorable fit show that G has the intended substantive meaning.

6.2 Three correlations exactly determine three higher-order loadings

With three standardized first-order factors and positive nonzero correlations,

$$\phi_{12} = \gamma_1\gamma_2, \qquad \phi_{13} = \gamma_1\gamma_3, \qquad \phi_{23} = \gamma_2\gamma_3.$$

Multiplying the first two equations and dividing by the third yields

$$\gamma_1^2 = \frac{\phi_{12}\phi_{13}}{\phi_{23}}, \qquad \gamma_2^2 = \frac{\phi_{12}\phi_{23}}{\phi_{13}}, \qquad \gamma_3^2 = \frac{\phi_{13}\phi_{23}}{\phi_{12}}. \tag{3}$$

Signs are chosen consistently with the factor orientation. Each ratio must be nonnegative and finite, and the standardized disturbance variances $1 - \gamma_j^2$ must be nonnegative. These are substantive parameter-space conditions, not algebraic decoration.

For the fitted Sample B M1-cross factor correlations, the calculation is:

Table 7

Closed-form higher-order coordinates from the fitted Sample B first-order factor correlations. These are standardized coordinates, not the SLE-generating population values.

Factor	Gamma	Disturbance
Planning	0.637	0.594
Strategy	0.637	0.595
Belonging	0.454	0.794

Thus $\hat{\gamma}_1 = 0.637$, $\hat{\gamma}_2 = 0.637$, and $\hat{\gamma}_3 = 0.454$. For example, the ratio for $\hat{\gamma}_1^2$ is 0.406 times 0.289, divided by 0.289, which equals 0.406.

The raw higher-order loadings printed by `lavaan` are on a different first-order disturbance metric when `std.lv = TRUE`; their standardized values are the axes that reproduce the fitted factor-correlation matrix above. Mixing the raw and standardized columns would break the closed-form check.

6.3 With three first-order factors, fit cannot choose between the forms

The correlated model estimates $m(m-1)/2$ first-order factor correlations. The one-factor higher-order form estimates m higher-order loadings. Their difference is

$$\frac{m(m-1)}{2} - m = \frac{m(m-3)}{2}. \tag{4}$$

For $m = 3$, the difference is zero. At a regular admissible solution, the three correlations and three higher-order loadings are alternative axes for the same information. The fitted records confirm this equivalence rather than treating matching fit as accidental.

Table 8
Sample B fit profiles for M1-cross, its equivalent higher-order representation, and the nonnested orthogonal bifactor rival. The extension fits are didactic post-confirmation analyses, not Sample-A-frozen rivals.

Model	T	df	p	RM-SEA	CI90	CFI	TLI	SRMR
M1_cross	48.475	49	0.494	0.000	[0.000, 0.034]	1.000	1.000	0.032
Higher_order	48.475	49	0.494	0.000	[0.000, 0.034]	1.000	1.000	0.032
Bifactor	69.258	42	0.005	0.042	[0.023, 0.060]	0.982	0.972	0.046

M1-cross and the higher-order form have 29 free covariance parameters, 49 df, the same fitted covariance matrix to numerical tolerance, and the same likelihood, AIC, and BIC. Therefore these data cannot decide which story is preferable. Interpretation must come from the intended construct representation and from evidence beyond the reproduced covariance.

For $m = 4$, six factor correlations are replaced by four higher-order loadings, so the higher-order model imposes two restrictions. The four-factor model introduced in Chapter 3 therefore supplies a testable case; three first-order factors do not. More first-order factors do not make the model automatically correct—they merely create restrictions that can be evaluated.

6.4 A bifactor model allocates item covariance directly

A standard bifactor model takes a different route. Every item loads directly on one general factor and on one assigned domain-specific factor:

$$Y_{ij} = \nu_j + \lambda_{jg} G_i + \lambda_{js} S_{is} + \epsilon_{ij}.$$

In the version fitted here, G, Planning-specific, Strategy-specific, and Belonging-specific are mutually orthogonal, residual covariances are zero, and each item has exactly one specific loading. A specific factor represents common variance among its assigned items after the general factor is held constant; it is not an item residual.

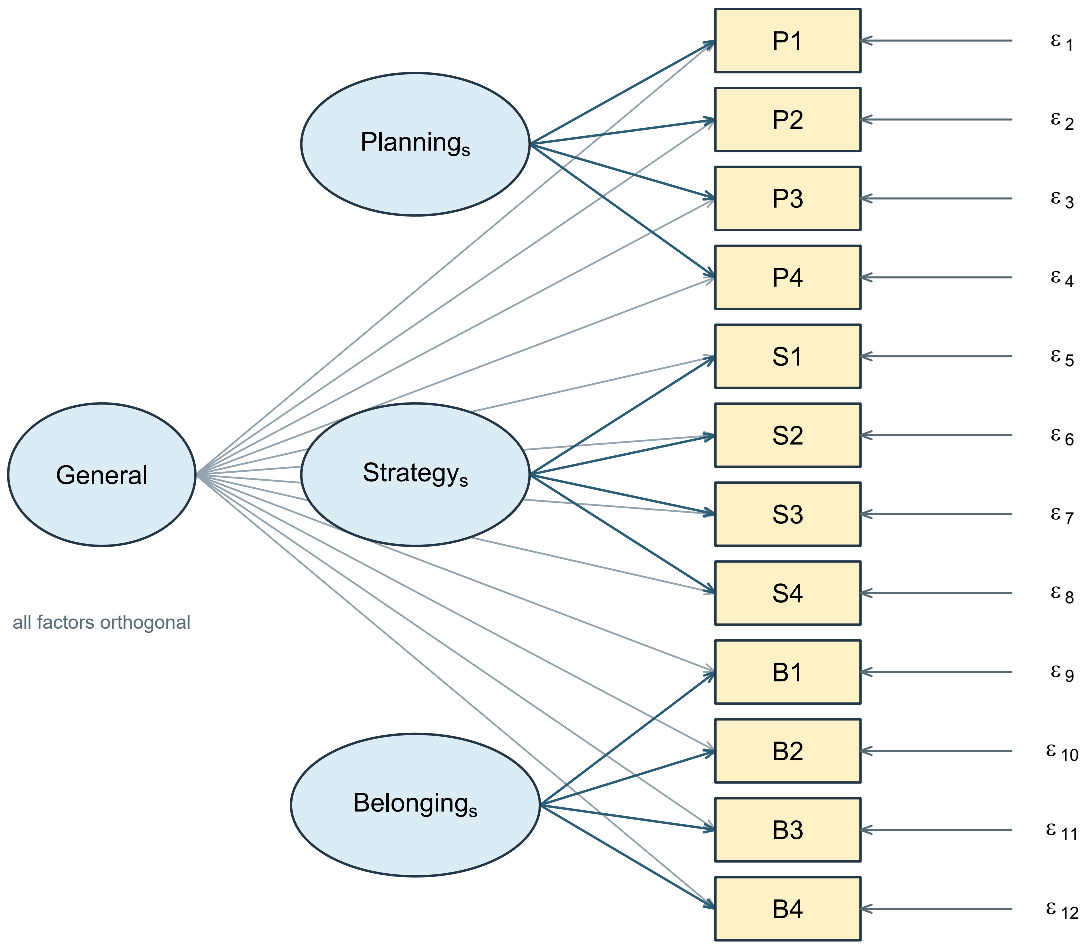

Figure 5: Orthogonal SLE bifactor model. General and the three domain-specific factors load directly on items; all four factors are mutually orthogonal and unshown loadings are zero.

The corresponding core syntax is explicit about orthogonality:

```
General =~ P1 + P2 + P3 + P4 + S1 + S2 + S3 + S4 +
           B1 + B2 + B3 + B4
Planning_s  =~ P1 + P2 + P3 + P4
Strategy_s  =~ S1 + S2 + S3 + S4
Belonging_s =~ B1 + B2 + B3 + B4
General ~~ 0*Planning_s + 0*Strategy_s + 0*Belonging_s
Planning_s ~~ 0*Strategy_s + 0*Belonging_s
Strategy_s ~~ 0*Belonging_s
```

Fixed-factor scaling leaves 24 loadings and 12 item residual variances, so $t = 36$ and $df = 78 - 36 = 42$. Relative to M1-cross, the bifactor adds a general loading for every item but also removes the three factor correlations and replaces the two M1 secondary loadings with exclusive domain-specific assignments. It is therefore a distinct, nonnested rival, not M1 plus twelve harmless arrows. An ordinary likelihood-ratio difference test between the two is not licensed.

Table 9

Fully standardized bifactor loadings in Sample B. Specific-factor coefficients describe covariance remaining after the orthogonal general factor; they need not all be large or positive.

Item	General	Factor	Specific	p specific
P1	0.322	Planning_s	0.701	0.000
P2	0.329	Planning_s	0.671	0.000
P3	0.262	Planning_s	0.637	0.000
P4	0.554	Planning_s	0.476	0.000
S1	0.672	Strategy_s	0.328	0.007
S2	0.628	Strategy_s	0.385	0.005
S3	0.611	Strategy_s	0.316	0.007
S4	0.889	Strategy_s	-0.180	0.570
B1	0.418	Belonging_s	0.776	0.000
B2	0.344	Belonging_s	0.716	0.000
B3	0.326	Belonging_s	0.521	0.000
B4	0.194	Belonging_s	0.411	0.000

The bifactor fit converges, has a full-rank local derivative matrix, and has positive estimated residual variances. Yet the Strategy-specific loading for S4 is -0.180, with $p = 0.570$. That weak negative coefficient is not a numerical failure, but it strains a simple claim that every item carries a clear positive domain-specific signal. The bifactor also has higher AIC and BIC than M1-cross. Greater flexibility and a familiar name do not guarantee a more useful representation.

6.5 The Schmid-Leiman transformation connects the two forms

The higher-order and bifactor models have been presented as rivals, but they are not unrelated. One is a restricted case of the other, and the algebra that connects them is worth doing, because it says exactly what the bifactor model adds.

Write the higher-order model with first-order loadings Λ, second-order loadings γ, first-order disturbance variances Ψ (diagonal) and item residuals Θ. Because the first-order factor covariance is $\gamma\gamma^{\mathsf{T}} + \Psi$, the implied covariance is

$$\begin{aligned}\Sigma &= \Lambda\left(\gamma\gamma^{\mathsf{T}} + \Psi\right)\Lambda^{\mathsf{T}} + \Theta \\ &= \underbrace{(\Lambda\gamma)(\Lambda\gamma)^{\mathsf{T}}}_{\text{general}} + \underbrace{(\Lambda\Psi^{1/2})(\Lambda\Psi^{1/2})^{\mathsf{T}}}_{\text{specific}} + \Theta.\end{aligned} \tag{5}$$

The second line *is* a bifactor model. This is the **Schmid-Leiman transformation** (Schmid & Leiman, 1957): a higher-order model rewritten as an orthogonal general factor with loadings $\Lambda\gamma$ and specific factors with loadings $\Lambda\Psi^{1/2}$. Nothing was estimated and nothing was tested; the same covariance structure was written a second way.

What the transformation exposes is a restriction, and it is worth stating precisely, because the general and specific loadings are built from the *same* first-order matrix. The general column is $\Lambda\gamma$ and the specific columns are $\Lambda\Psi^{1/2}$, so both are determined by Λ together with m values of γ. A standard bifactor model estimates its general and specific loadings freely and is under no such obligation.

For an item that loads on one first-order factor only, the restriction takes a particularly readable form. Item j in domain k receives a general loading $\lambda_{jk}\gamma_k$ and a single specific loading $\lambda_{jk}\psi_k^{1/2}$, so their ratio

$$\frac{\lambda_{jk}\gamma_k}{\lambda_{jk}\psi_k^{1/2}} = \frac{\gamma_k}{\psi_k^{1/2}} \tag{6}$$

does not depend on j: every simple-structure item in a domain divides its common variance between general and specific in the same proportion.

An item with two first-order loadings behaves differently, and the SLE specification contains two such items. Because P4 loads on both Planning and Strategy, its general loading sums both pathways, $\lambda_{41}\gamma_1 + \lambda_{42}\gamma_2$, and it acquires a *separate* specific loading on each of the two domains. Table 10 gives the exact transformation for the model of Section 8, computed from the full loading matrix rather than from primary loadings alone.

Table 10
Schmid-Leiman transformation of the higher-order reparameterization of the generating model. General loadings sum over every nonzero first-order pathway; P4 and S4 each receive two specific loadings. The constant ratio holds for simple-structure items only.

Item	General	Planning s	Strategy s	Belonging s	Ratio
P1	0.468	0.624			0.750
P2	0.444	0.592			0.750
P3	0.420	0.560			0.750
P4	0.554	0.440	0.229		–
S1	0.532		0.543		0.980
S2	0.504		0.514		0.980
S3	0.476		0.486		0.980
S4	0.535		0.393	0.260	–
B1	0.400			0.693	0.577
B2	0.370			0.641	0.577
B3	0.335			0.580	0.577
B4	0.250			0.433	0.577

The two rows without a constant ratio are the informative ones. They show that the Schmid-Leiman image of a higher-order model inherits the *whole* first-order loading pattern, cross-loadings included. A higher-order model is therefore nested within the bifactor model that shares that pattern - the one in which P4 and S4 each keep two specific loadings - and the difference in fit between those two tests proportionality rather than the existence of a general factor.

That is not the bifactor model fitted in Section 6.4. There, each item was assigned to exactly one specific factor, which is a different set of restrictions in a different place; as noted, it is a nonnested rival to M1-cross, and it remains one. Nesting holds between a higher-order model and its own Schmid-Leiman image, not between a higher-order model and any bifactor model that happens to have the same number of factors. Comparing the wrong pair and reading the difference as a test of proportionality is an easy and consequential mistake.

This reframes the choice. A general factor is not established by fitting a bifactor model and finding it fits; it is distinguishable only when the data depart from proportionality in a stable, interpretable way (Chen, 2026). A bifactor solution that satisfies Equation 6 closely is telling you that a higher-order account would serve, with fewer parameters and a clearer substantive story.

6.6 Where specific factors come from

Nothing above says what a specific factor *is*, and the answer is often not "a narrower trait". Two origins are common enough to have names, and both motivate a bifactor specification more convincingly than a wish for a general factor does.

A **method effect** is common variance among items that share a format rather than a content domain: reverse-keyed items, a distinct response scale, items answered by a different informant. A **testlet effect** is common variance among items attached to the same stimulus - the comprehension questions following one reading passage, the items scored from a single case vignette, a block of tasks sharing one apparatus. In both cases the extra dependence is a property of how the items were administered, and a model that ignores it pushes that dependence into the residuals, where a diagonal Θ forbids it and the strain reappears as local misfit.

Read this way, a specific factor is the multi-item generalization of a residual covariance. Two items sharing a passage could be modelled with one freed θ_{jk}; four items sharing it are more parsimoniously modelled with a small orthogonal factor. That is why testlet and method structures are where bifactor models are least controversial: the grouping is known in advance from the design, not discovered in the fitted output, and the specific factors are not competing with the general factor for a substantive interpretation.

The same structure recurs across quite different literatures - testlets in educational measurement, method factors in multitrait-multimethod designs, wording effects on balanced attitude scales, and item-bundle dependence in health and personality inventories - and the modeling question is the same each time: is the extra dependence a nuisance to be absorbed, or an effect to be reported? The answer belongs to the design and the intended use of the scores, not to the fit statistics.

6.7 Omega-hierarchical answers a score-specific question

For declared score weights $\mathbf{w}$, the share of observed score variance attributable specifically to the bifactor general factor is

$$\omega_h(\mathbf{w}) = \frac{(\mathbf{w}^\top \lambda_g)^2 \phi_{gg}}{\mathbf{w}^\top \Sigma \mathbf{w}}. \tag{7}$$

For the unit-weighted sum of all 12 items, the fitted bifactor gives $\omega_h = 0.612$ and total model-based omega $= 0.884$. Thus about 0.693 of the score's modeled reliable variance is attributed to the general factor. The remainder is modeled reliable specific-factor variance, not automatically error.

These coefficients are conditional on this model, these weights, and this sample. A substantial ω_h does not prove essential unidimensionality, validate the score interpretation, or erase the weak specific-loading pattern. It answers a narrower variance-decomposition question.

7 Diagnose, revise, and validate

7.1 MI and EPC are conditional local diagnostics

A modification index (Sörbom, 1989) approximates the improvement in the model statistic if one fixed parameter were released and the model re-estimated. The expected parameter change, EPC, approximates the parameter's change under that local release. The standardized EPC expresses a corresponding rescaled magnitude, with the exact scale depending on the parameter type.

Read them together with the residual pattern, parameter meaning, and analytic history:

$$(\mathrm{MI}, \mathrm{EPC}, \text{standardized EPC}, \text{residual pattern}, \text{meaning}, \text{status}).$$

An MI is not the exact LR change, a posterior probability of a missing path, or a guarantee that the proposed parameter is substantively real. Searching many MIs and reporting only the largest ordinary p value does not retain the interpretation of a single prespecified test. The official modification-index guide also distinguishes the different standardized EPC columns.

Table 11
Largest one-parameter diagnostics for M1 in Sample B. No cutoff is used as an automatic release rule.

Left	Relation	Right	MI	EPC	Std.EPC
Planning	is measured by	B4	4.949	-1.300	-0.126
P4	covaries with	S3	4.252	-6.996	-0.138
Planning	is measured by	S1	3.741	-1.064	-0.106
S3	covaries with	B3	3.671	6.010	0.115
S3	covaries with	B4	3.608	-8.506	-0.111
S2	covaries with	B3	3.125	-3.977	-0.108

The top suggestion is a negative B4 loading on Planning, with MI 4.949 and EPC -1.300. It aligns with some negative cross-domain residuals involving B4. But institutional connection wording does not offer a compelling Planning mechanism. The numerical clue is therefore weaker than a justified measurement revision.

By contrast, the D2 example's missing F1 loading on Y3 was already one of the registry's specified alternatives. A planned rival and a newly discovered relation may be represented by the same software operation, yet have different analytic status.

7.2 A deliberately tentative one-path sensitivity

For teaching, we retain one tentative model, **R1-B4**, which frees Planning's loading on B4 while leaving all other M1 restrictions unchanged. It has 30 free parameters and 48 df. This is a transparent sensitivity demonstration, not a recommendation to modify a favorable model whenever an MI is available.

The revision's syntax, its stated limitations and its parent model were all fixed before Sample C entered any model fit or diagnostic analysis; the provenance check had touched it only to confirm the released file matches the declared population. Only this one diagnostic revision is fitted. We do not chase additional MIs, remove items, select a new seed, or invent an explanation after seeing C.

In Sample B, the actual decrease in the model statistic is 5.040 on one df, close to but not identical to the MI. The estimated added loading is -1.318, with 95% interval [-2.468, -0.169]. Its conventional same-sample LR p value is 0.025. Because this path was selected from the diagnostics, that value is descriptive of the refit, not an unadjusted confirmatory discovery.

7.3 The held-back retest does not support the added path

Table 12

One frozen revision across its development and revalidation records. Compare models within B or within C; do not compare likelihood values across samples.

Model	T	df	p	RMSEA	CI90	CFI	TLI	SRMR
B_M1	48.475	49	0.494	0.000	[0.000, 0.034]	1.000	1.000	0.032
B_R1	43.435	48	0.660	0.000	[0.000, 0.029]	1.000	1.004	0.025
C_M1	43.794	49	0.684	0.000	[0.000, 0.028]	1.000	1.004	0.026
C_R1	43.750	48	0.648	0.000	[0.000, 0.029]	1.000	1.003	0.026

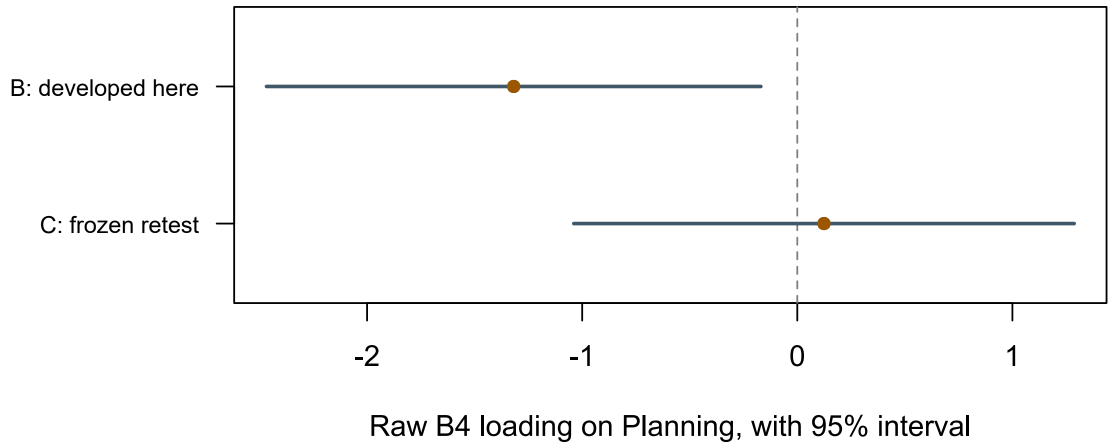

Figure 6: The added B4-on-Planning loading in the sample that suggested it and in the untouched retest. Intervals are 95 percent conventional parameter intervals.

In Sample C the added coefficient is 0.124, with interval [-1.039, 1.287]. The LR improvement is only 0.043 on one df, with $p = 0.835$. The candidate relation does not receive independent support in this retest. We retain M1-cross as the course's working measurement model and do not promote R1 to the final specification.

This is not proof that the population coefficient is exactly zero. It is a bounded decision based on weak content justification, a data-selected improvement in B, and no corresponding evidence in C. The failed sensitivity remains in the report rather than disappearing from the analytic history.

7.4 What validation does and does not fix

A held-out sample protects a specified retest from using the same sampling fluctuations that suggested the change. It does not repair poor content coverage, turn simulation into external validity, or justify unlimited searches in the held-out record. Once C is used to choose further modifications, it becomes development evidence for those decisions.

Validation here means re-estimating and evaluating a fixed model structure in independent observations from the same synthetic population. It does not mean holding every B estimate fixed as a predictive parameter value. That would be a different, more stringent out-of-sample prediction question requiring its own objective and evaluation protocol.

Stage	Decision	Evidence role
A	Retain three factors and two secondary loadings	Development
Before B	Fix M0–M4 and the M1 target	Specification settled
B, Chapter 3	Estimate M1 and check its solution	Confirmation-record estimation
B, Chapter 4	Evaluate planned models; inspect local diagnostics	Planned evaluation plus labeled exploration
Before C	Fix the tentative R1-B4 without further changes	Retest specification settled
C	Evaluate M1 and R1; do not search again	Independent synthetic validation
Report	Retain M1; disclose unsupported R1	Bounded conclusion

8 Evaluate the fixed model against the known population

8.1 Reveal the generating feature only after the B/C sequence

The SLE records are synthetic, so the generating covariance is available to the course author. It was deliberately withheld while the fixed B/C sequence was conducted. We now use it retrospectively to ask what the working M1-cross representation misses. This comparison cannot retroactively make a diagnostic planned, and it does not authorize another model search.

The generating residual covariance between B1 and B2 is 0.120 on the standardized item scale (14.400 on the raw covariance scale). M1-cross fixes that relation to zero. A simple four-indicator tetrad makes the restriction visible. In the generating correlation matrix,

$$r_{12}r_{34} - r_{13}r_{24} = .712(.335) - .536(.370) = .0402,$$

whereas the fitted one-factor Belonging block imposes zero, apart from numerical tolerance. The nonzero population tetrad identifies a specific covariance restriction that global indices summarize but cannot name.

Now that the generator is revealed, its first-order correlations also give a clean higher-order calculation:

$$\gamma_1^2 = \frac{.42(.30)}{.35} = .36, \qquad \gamma = (.60, .70, .50)^\mathsf{T}, \qquad \mathrm{diag}(\Psi_h) = (.64, .51, .75).$$

These are **generating-population axes**, not the Sample B estimates in Section 6. Because three factor correlations can be reparameterized this way, the calculation does not show that a higher-order causal process generated the data; the documented SLE generator simply used a correlated factor covariance matrix.

8.2 Population fitting separates displacement from sampling fluctuation

Fit the unchanged M1-cross model directly to the exact generating covariance matrix. The optimizer then finds the population-ML, or pseudo-true, axes under that misspecified model. No random sample is involved in the following loading displacement:

Table 13

Belonging loadings in the generating model and at the population-ML optimum under M1-cross. Displacement is deterministic under these two specifications; it is not inferred from one realized sample.

Item	Generating	Pseudo true	Displacement
B1	0.800	0.868	0.068
B2	0.740	0.811	0.071
B3	0.670	0.625	-0.045
B4	0.500	0.471	-0.029

The two indicators sharing the omitted residual relation have higher pseudo-true primary loadings; the other two have lower loadings. This is a systematic consequence of fitting the stated restricted model to the stated population covariance. By contrast, comparing two estimates in Sample B can show sensitivity to a specification choice but cannot by itself establish estimator bias.

8.3 Population discrepancy is not the sample RMSEA formula

For population covariance Σ_0 and its pseudo-true fitted covariance Σ^*, define

$$F_0 = log|\Sigma^*| + \mathrm{tr}(\Sigma_0 \Sigma^{*-1}) - \log|\Sigma_0| - p.$$

The comparison gives $F_0 = 0.01159907$. Population RMSEA is therefore

$$\mathrm{RMSEA}_0 = \sqrt{F_0/df} = \sqrt{0.01159907/49} = 0.015.$$

Table 14

Population discrepancy measures versus sample-index formulas applied to the same population covariance with $N = 360$. The latter are not observed-sample evidence.

Quantity	Population definition	Sample formula at N360
Model statistic	NA	4.176
df	49.000	49.000
RMSEA	0.015	0.000
CFI	0.997	1.000
SRMR	0.011	0.011

Applying the ordinary finite-sample formulas at $N = 360$ to population input produces RMSEA .000 and CFI 1.000 because their degree-of-freedom corrections truncate the small discrepancy. Those numbers do **not** make the model true: the population discrepancy, the tetrad, and the loading displacement remain nonzero. Even exact knowledge of the population covariance does not turn a global index into a certificate of specification.

9 Interpret parameters, reliability, and score boundaries

9.1 Parameters still depend on the model

Chapter 3's parameter intervals remain conditional on M1's restrictions and ML assumptions. The favorable fit profile makes ordinary interpretation more defensible than it would be under severe misspecification, but it does not remove that conditionality. Residual inspection also shows that individual relations are not reproduced perfectly.

B4's relatively weak primary loading can reflect a broader or different content emphasis. Removing it would alter the domain represented by Belonging. It would also change the observed outcome set and invalidate a direct AIC/BIC comparison with the twelve-item fit. "Higher loading after deletion" is not a complete scale-development rationale.

Factor correlations are estimated associations, not reliability coefficients or directional effects. A correlation close to one can raise a question about empirical separation, but a universal cutoff cannot replace checking uncertainty, item content, and alternative model structures. The present three correlations are moderate positive relations in the fixed-factor metric.

9.2 Define a score before calculating reliability

For a continuous composite

$$c_i = \mathbf{w}^\mathsf{T}\mathbf{y}_i,$$

its model-implied variance is $\mathbf{w}^\mathsf{T}\Sigma\mathbf{w}$. The proportion attributed to all modeled common factors is

$$\omega(\mathbf{w}) = \frac{\mathbf{w}^\mathsf{T}\Lambda\Phi\Lambda^\mathsf{T}\mathbf{w}}{\mathbf{w}^\mathsf{T}\Sigma\mathbf{w}}.$$

The numerator is the variance of the model's common part for that weighting rule. The denominator also includes $\mathbf{w}^\mathsf{T}\Theta\mathbf{w}$. This expression retains cross-loadings, factor correlations, and any modeled residual covariances. A shortcut based only on selected loadings may omit essential terms.

For the course example, define four scores on the *raw process-index scale*: the mean of P1–P4, the mean of S1–S4, the mean of B1–B4, and the mean of all twelve indicators. Weights are .25 on the relevant four items and zero elsewhere for a subscale, or 1/12 on all items for the total. A standardized-item mean would be a different weighting rule and requires a separately defined calculation.

Table 15

Model-conditional common-variance proportions for explicitly defined raw-item composites under M1 in Sample B. These are point estimates, not validation thresholds.

Composite	Common variance proportion
Planning mean	0.811
Strategy mean	0.812
Belonging mean	0.792
All-item mean	0.881

The all-item mean has a larger common-variance proportion than the individual subscale means, but that does not make it unidimensional. Its numerator combines common variance from three correlated factors. Similarly, the Planning and Strategy means include their retained secondary-loading contributions. Their values are not proportions attributable exclusively to the intended named factor.

Reliability concerns a specified score under a specified model and population. It does not establish dimensionality, content coverage, fairness, predictive usefulness, or invariance across groups. The reported estimates also have sampling uncertainty. A bootstrap interval would require a declared resampling design, model-refitting checks, and an explicit treatment of failed or improper solutions; it would not account automatically for every earlier model-selection decision.

9.3 A small calculation exposes the assumptions

For a unit-weighted sum of three indicators loading only on a unit-variance factor, with loadings $.8, .7, .6$ and uncorrelated residual variances $.36, .51, .64$,

$$\omega = \frac{(.8 + .7 + .6)^2}{(.8 + .7 + .6)^2 + .36 + .51 + .64} = \frac{4.41}{5.92} \approx .745.$$

If the first two residuals instead have covariance .10, the denominator increases by .20, because both symmetric covariance entries enter the sum variance. The common numerator is unchanged. Ignoring that residual relation would overstate the common-variance proportion.

The signs also matter. Replacing loadings by absolute values silently changes the implied score unless the indicator coding and weights are correspondingly changed. Reliability is not improved by hiding opposing contributions inside an absolute-value formula.

AVE can summarize the average communality for a simple, fully-standardized single-factor item set. It is not interchangeable with reliability of a weighted composite, and the SLE cross-loading subscales do not satisfy that simple interpretation without qualification. Coefficient H requires its own optimal-weighting conditions and is left to the optional note.

9.4 Factor scores are predictions, not observed latent values

A factor score uses a model-dependent weighting matrix:

$$\hat{\eta}_i = \mathbf{A}(\mathbf{y}_i - \hat{\mu})$$

under the zero latent-location convention. The regression score targets a conditional mean under a normal model and typically shrinks uncertain individuals toward the latent mean. Bartlett's score uses a different weighting criterion, emphasizing residual-variance weighting. Both use estimated parameters and can differ for the same person under the same fitted covariance model.

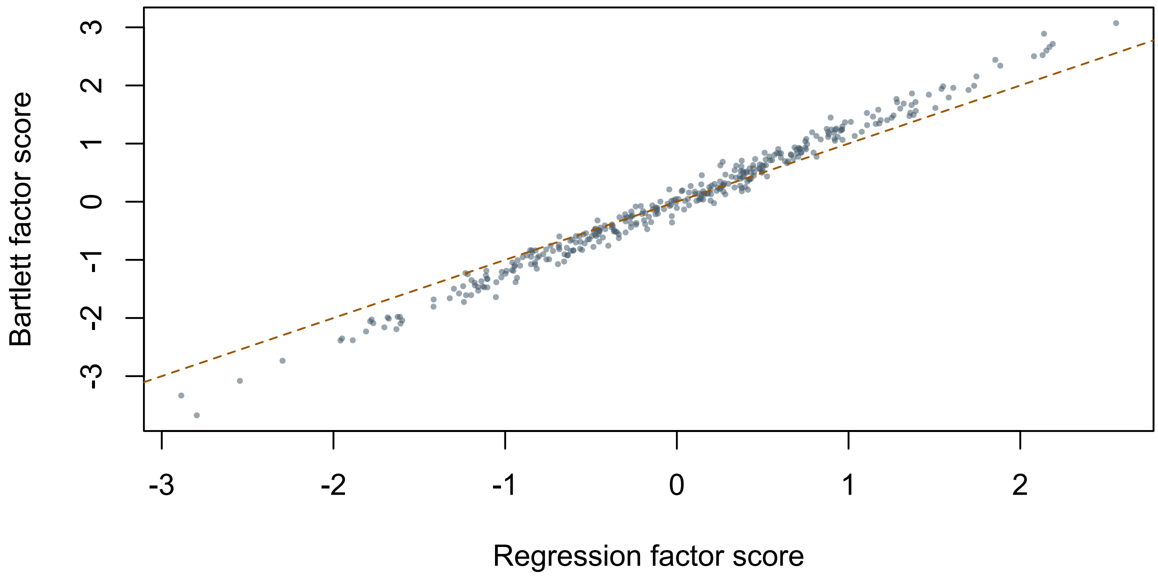

Figure 7: Two ways of scoring Planning in the same synthetic Sample B. Agreement is not identity and neither axis is an observed true factor value.

Under the fitted normal model with parameters treated as known, the conditional SDs of the three latent factors given the indicators are approximately 0.428, 0.417, 0.365. The factors have unit marginal variance, so these are not zero uncertainties. They also omit parameter-estimation and model-selection uncertainty. A numerical score is a prediction, not the elimination of the measurement problem.

Exporting estimated factor scores into an ordinary regression and treating them as perfectly measured covariates can change the target and distort inference. For many substantive questions, a joint latent SEM is the appropriate next representation, provided its measurement and structural assumptions are justified. Scores can still be useful for clearly stated descriptive purposes; their method, scale, and uncertainty should accompany them.

10 Report a reproducible CFA evaluation

10.1 Report in the order of dependence

A report should state the sample and data role, indicators and coding, model restrictions, factor metric, estimator and likelihood convention, mean structure, and missing-data/design handling before presenting fit. Follow with convergence and admissibility, global and local evidence, planned comparisons, any data-developed changes, parameter interpretation, and the limits of the conclusion.

A possible compact report for this record is:

> We evaluated a previously fixed twelve-indicator, three-factor model in a simulated confirmation sample of 360 complete continuous records. The model retained two specified secondary loadings, free factor correlations, unit factor variances, and diagonal residual covariance. Conventional normal ML was used without a mean structure. The fit converged without warnings and was admissible. M1-cross had $\chi^2(49) = 48.47$, $p = 0.494$, RMSEA = 0.000 (90% CI [0.000, 0.034]), CFI = 1.000, TLI = 1.000, and SRMR = 0.032. Planned restrictions removing the secondary loadings or forcing orthogonality were opposed by compatible ML difference tests. Local residuals involving B4 motivated one tentative sensitivity model. Its added loading did not receive support in an independent held-back retest of 360 simulated records, so the primary model was retained. Reliability estimates refer to declared raw-item composites and do not establish unidimensionality or general validity.

The report includes a favorable conclusion about a working covariance representation without claiming the factors have been directly observed or the measurement theory uniquely confirmed. It also retains the unsuccessful sensitivity analysis, a material part of what was done.

10.2 Preserve reproducibility without overwhelming the main text

Archive the exact syntax, data manifest and codebook, parameter tables, fitted objects, comparison eligibility checks, software versions, and fixed revision record. The main text need not print every diagnostic row, but its selected rows must be traceable and must not conceal an improper fit or unsuccessful comparison.

The canonical analysis reconstructs the ML statistic, RMSEA and its interval, CFI, TLI, SRMR, AIC, BIC, and eligible LR differences from the same fitted family. Reproduction is an arithmetic and provenance check; it does not replace substantive review. Public chapter renders have no dependency on private teaching files or archived data sets.

11 Bridge from CFA to structural equation modeling

11.1 Add structural restrictions without discarding measurement

The measurement equation remains $\mathbf{y} = \nu + \Lambda\eta + \epsilon$. A later structural model can write

$$\eta = \alpha + \mathbf{B}\eta + \Gamma\xi + \zeta,$$

where $\mathbf{B}$ contains latent structural paths, Γ links exogenous predictors, and ζ is a structural disturbance. If $\mathbf{I} - \mathbf{B}$ is invertible,

$$E(\eta) = (\mathbf{I} - \mathbf{B})^{-1}\{\alpha + \Gamma E(\xi)\}.$$

Thus α remains the latent *intercept*, not generally the marginal latent mean. In the earlier CFA, $\mathbf{B} = \mathbf{0}$ and there were no predictor terms, so the two coincided. No new intercept symbol is required when structural paths arrive.

Let $\Psi = \mathrm{Var}(\zeta)$. If predictors and disturbances are uncorrelated, the total latent covariance is

$$\Phi = (\mathbf{I} - \mathbf{B})^{-1}\{\Gamma\,\mathrm{Var}(\xi)\Gamma^{\mathsf{T}} + \Psi\}(\mathbf{I} - \mathbf{B})^{-\mathsf{T}}.$$

Φ still means total latent covariance; Ψ now means structural disturbance covariance; Θ remains indicator residual covariance. Keeping these roles distinct prevents a notation change from hiding a change in the statistical model.

12 Summary and reading bridge

CFA evaluation is not a vote among fit-index cutoffs. The model test addresses exact covariance restrictions; RMSEA describes approximation per df with uncertainty; CFI and TLI use a baseline; SRMR summarizes standardized covariance residuals. Local patterns and parameter meaning remain necessary after those summaries are read.

Planned comparisons require exact nesting and compatible likelihoods. Information criteria require comparable observations and outcomes. Score, Wald, and LR tests take different views of a restriction and need not agree numerically. Modification indices are local approximations whose interpretation changes when selected from a search.

The SLE record supports retaining M1-cross as a working measurement model within the synthetic course example. A tentative B4 cross-loading improved its development record but failed to receive support in the untouched retest. Reliability calculations remain conditional on the model and weighting rule; factor scores remain uncertain predictions.

The fitted three-factor higher-order form is an admissible reparameterization of M1-cross, so identical fit cannot choose between their interpretations. The orthogonal bifactor model is a nonnested rival: it fits reasonably but less well by AIC/BIC here and contains a weak negative specific loading. Omega-hierarchical describes the declared all-item score's general-factor variance; it does not certify unidimensionality. Finally, the known-population comparison reveals a nonzero omitted B1–B2 residual relation, loading displacement, and population discrepancy even though finite-sample fit formulas applied at $N = 360$ look perfect.

For further study, return to Bollen (1989) and Brown (2006) for the measurement-model framework, and MacCallum et al. (1996) for the connection between fit uncertainty and power.

Read software tutorials alongside their fitted-object definitions, not as authority for universal cutoffs. The next topic is how structural restrictions change the model-implied latent means and covariances while retaining the measurement system.

Exercises

1. **Reconstruct ML.** For $\mathbf{S} = \begin{bmatrix} 1 & .5 \\ .5 & 1 \end{bmatrix}$ and $\Sigma = \begin{bmatrix} 1 & .3 \\ .3 & 1 \end{bmatrix}$, calculate F_{ML} and NF_{ML} for $N = 200$. Explain why a chi-square reference still requires the model's df and regularity conditions, not only this arithmetic.
2. **RMSEA and uncertainty.** Compute RMSEA under the N convention for $T = 60$, $df = 40$, $N = 400$, and for $T = 35$ with the same df and N. Explain why the second result does not imply zero residuals. What does a wide 90% interval add to interpretation?
3. **Baseline dependence.** With $T_M = 60$, $df_M = 40$, compare CFI when $(T_B, df_B) = (500, 60)$ versus $(100, 60)$. Explain why the same target-model statistic can yield different incremental fit and why neither result establishes construct truth.
4. **Establish nesting.** Show the exact restrictions producing D1, D2, and D4 from D3. Use the printed statistics to calculate D4 versus D3's LR difference and df. Explain why D1 and D2 cannot be compared by the same ordinary subtraction test.
5. **Evaluate the tentative revision.** Combine B4's wording, residual pattern, MI/EPC, B refit, and C retest. Explain the decision to retain M1 without claiming that C proves the added loading is exactly zero. What would be lost by omitting R1 from the report?
6. **Invert three factor correlations.** Starting from the fitted Sample B correlations in Section 6, recover the three standardized higher-order loadings and disturbance variances. State the sign, denominator, and admissibility conditions. Explain why the higher-order form has the same df and fit as M1-cross, and why a four-first-order-factor version would impose two restrictions.
7. **Compare a bifactor rival.** Count the 36 free covariance parameters and 42 df in the displayed bifactor model. Explain why it is not nested in M1-cross, interpret the weak Strategy-specific S4 loading, and compare their fit profiles without claiming that a general factor has been proved.
8. **Read the population comparison.** Use the displayed tetrad and loading-displacement table to identify the omitted restriction. Explain why the population RMSEA is nonzero while the finite-N sample formulas applied to the population covariance return RMSEA .000 and CFI 1.000, and why neither result is a specification certificate.

Appendix: Interval and score calculations

Inverting the noncentral chi-square reference

Let $G(t; d, \delta)$ be the noncentral chi-square CDF with df d and noncentrality δ. For a two-sided 90% interval, solve

$$G(T_M; df_M, \delta_L) = .95, \qquad G(T_M; df_M, \delta_U) = .05,$$

with a lower boundary of zero when an interior solution does not exist. Transform the limits using $\sqrt{\delta/(N df_M)}$ under this chapter's convention. The shared numerical helper brackets the roots and reproduces the package intervals for every displayed model. The numerical inversion is optional; interpreting why a point estimate needs uncertainty is core.

Regression and Bartlett weights

With a jointly normal model, zero latent mean, and fitted parameters temporarily treated as known, the regression weights are

$$\mathbf{A}_R = \Phi \Lambda^\mathsf{T} \Sigma^{-1},$$

and the conditional covariance is

$$\mathrm{Var}(\eta \mid \mathbf{y}) = \Phi - \Phi \Lambda^\mathsf{T} \Sigma^{-1} \Lambda \Phi.$$

For nonsingular residual covariance and a full-column-rank loading matrix, Bartlett weights are

$$\mathbf{A}_B = (\Lambda^\mathsf{T} \Theta^{-1} \Lambda)^{-1} \Lambda^\mathsf{T} \Theta^{-1}.$$

These formulas clarify why scoring depends on more than selecting the largest loadings. With nonzero latent means, add the appropriate latent mean and center indicators at their implied means. Parameter uncertainty and model uncertainty require additional treatment; the conditional covariance above does not include them.

For a factor a, model-based regression-score determinacy is the correlation between the latent factor and its regression prediction. Under the stated model its square is $1 - \mathrm{Var}(\eta_a \mid \mathbf{y})/\phi_{aa}$. High determinacy is not proof of a correct measurement model, and it does not eliminate prediction error for individuals.

Coefficient H under a restrictive one-factor model

For a single unit-variance factor with diagonal positive residual covariance, let $a = \lambda^{\mathsf{T}}\Theta^{-1}\lambda$. The maximum common-variance proportion among linear composites is

$$H = \frac{a}{1+a}.$$

With fully standardized simple indicators, $a = \sum_j \lambda_j^2/(1-\lambda_j^2)$. This is an optimal-weighting result under specific assumptions, not reliability of every unit-weighted score and not a general formula for a multifactor cross-loading subscale. It should not be attached to the SLE factors by ignoring their other loadings.

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